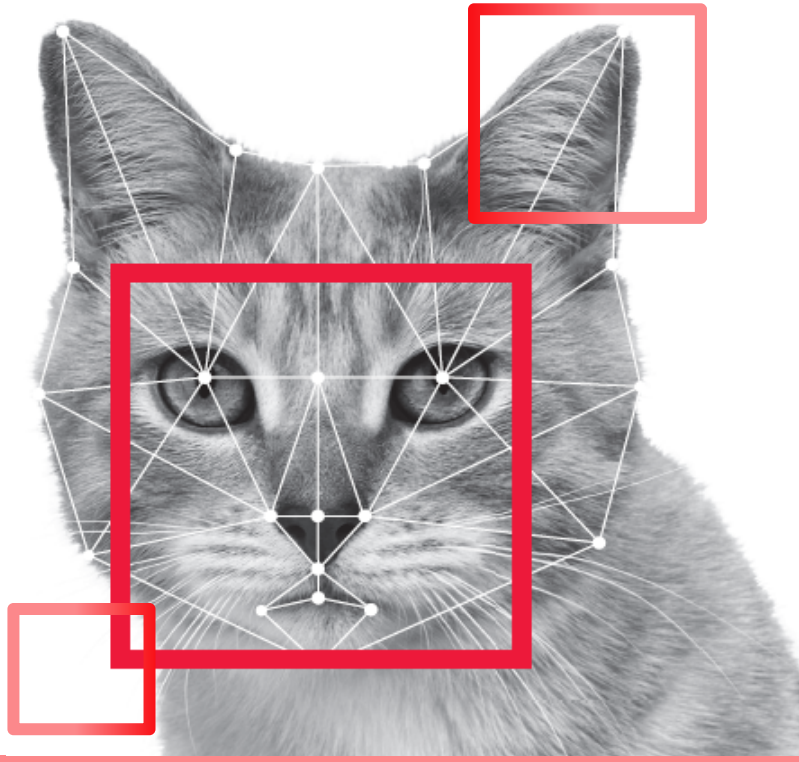
COMPUTER VISION 컴퓨터 비전

기본 개념부터 최신 모바일 응용 예까지

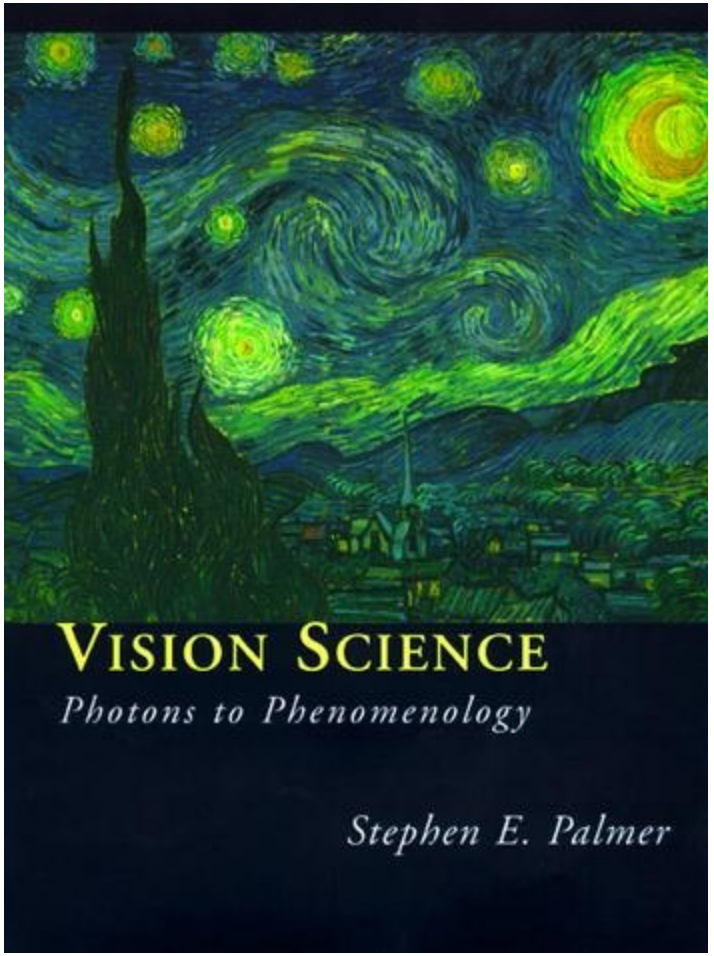


2장. 칼라

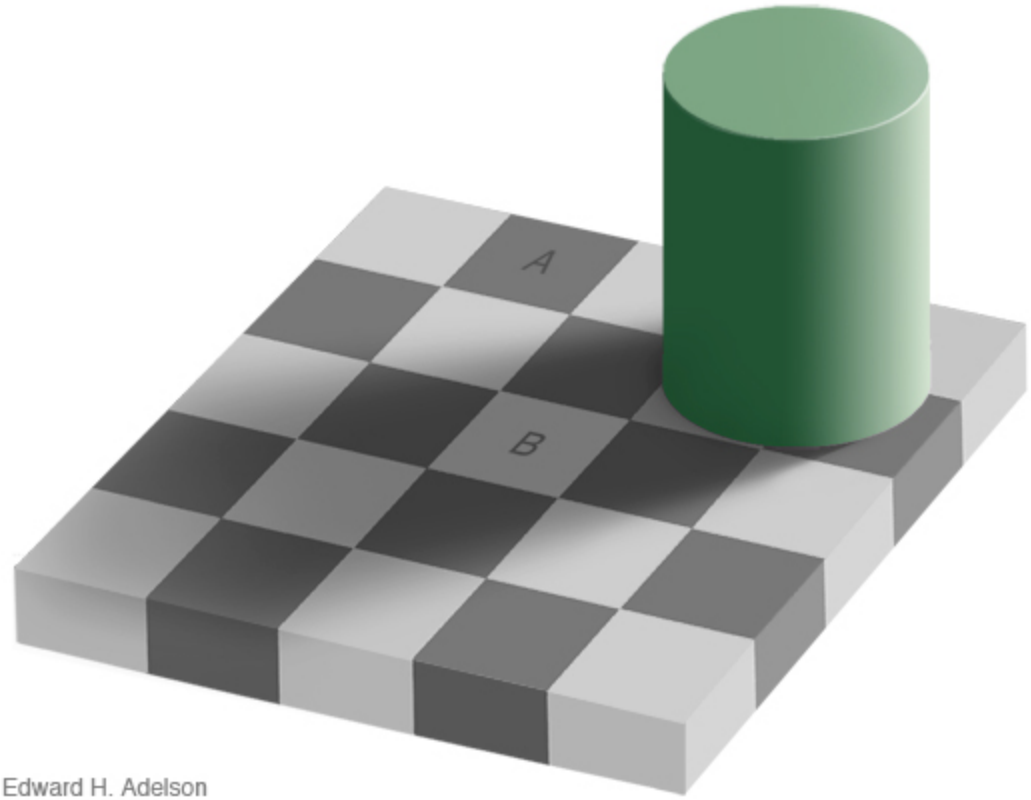
각 절에서 다루는 내용

1. 컬러

2. Color is the result of interaction between physical light in the environment and our visual system
3. Color is a psychological property of our visual experiences when we look at objects and lights, *not* a physical property of those objects or lights
(S. Palmer, *Vision Science: Photons to Phenomenology*)



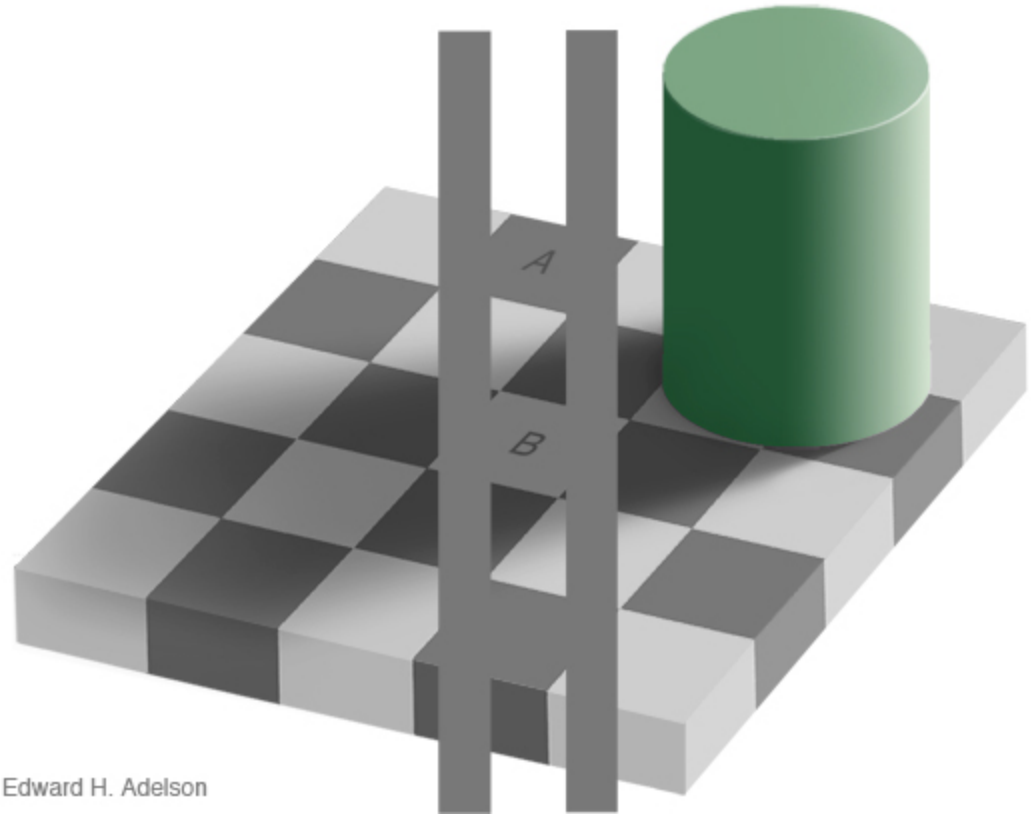
Checker shadow illusion



Edward H. Adelson

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

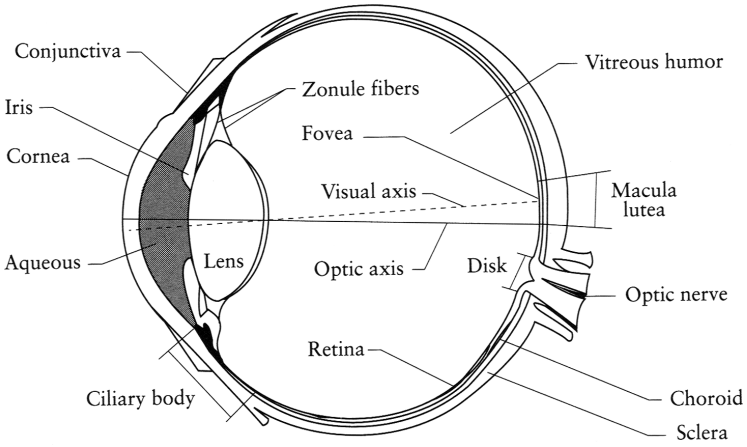
Checker shadow illusion

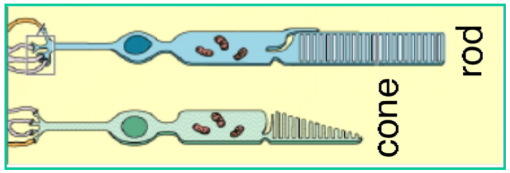


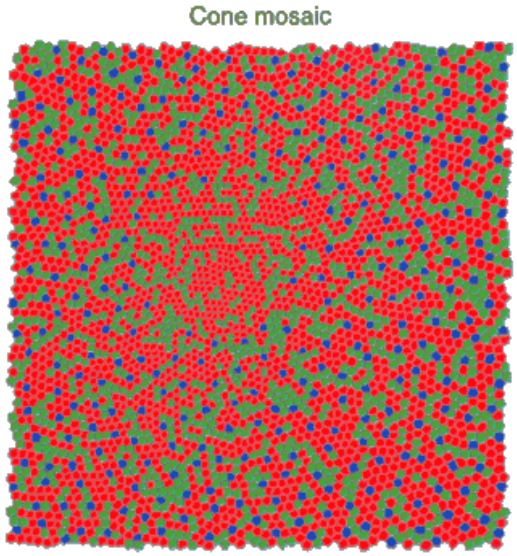
- Possible explanations
  - Simultaneous contrast
  - Reflectance edges vs. illumination edges

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

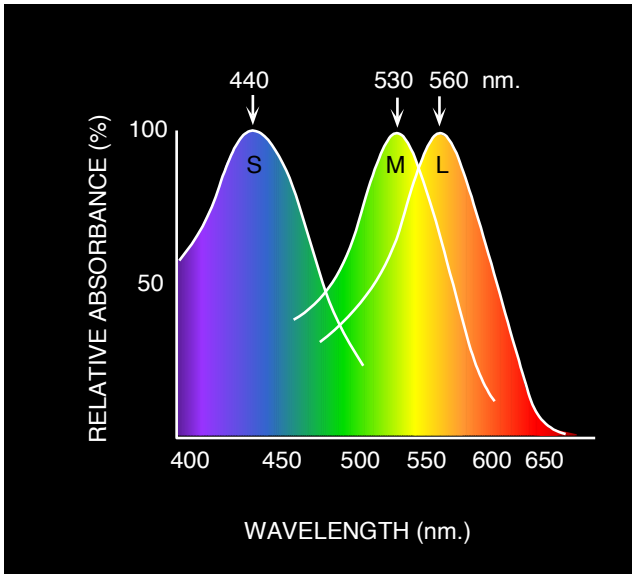
The Eye







R G B
가장 많다

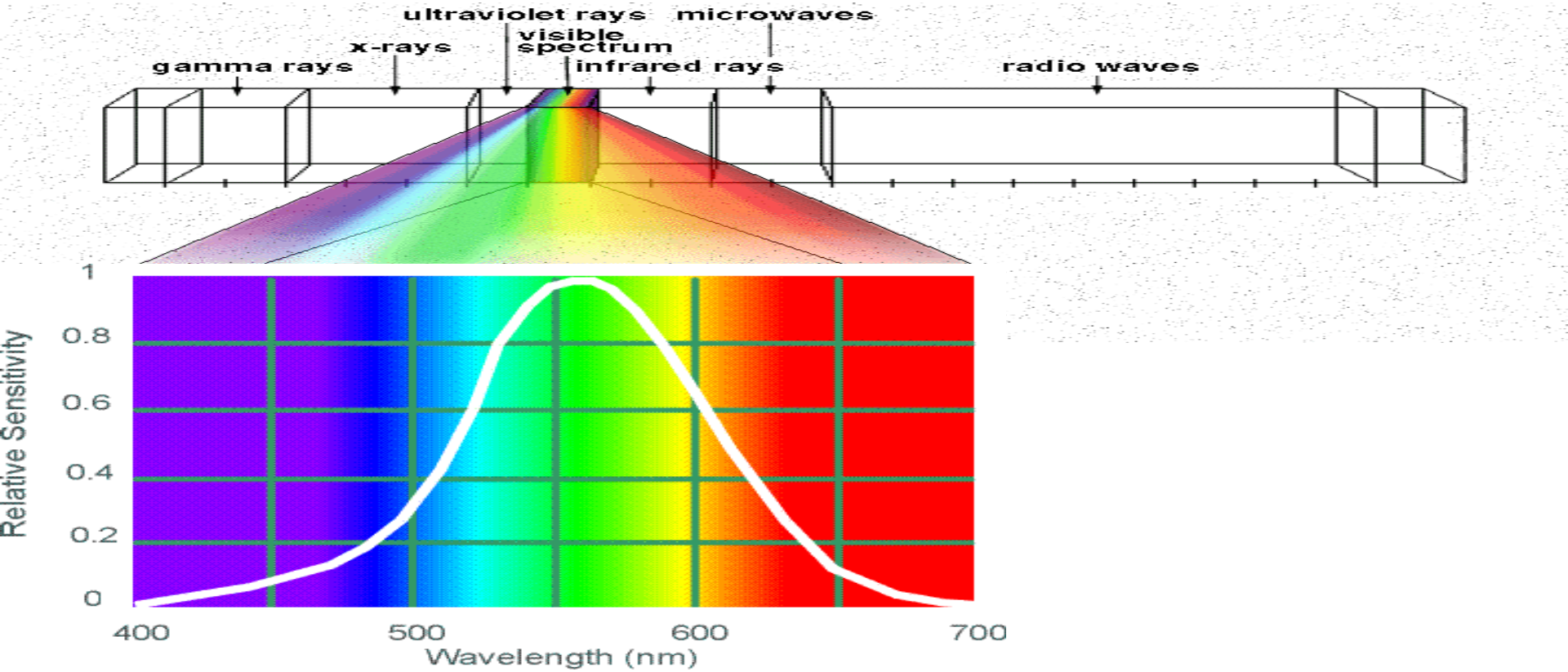


- Ratio of L to M to S cones: approx. 10:5:1
- Almost no S cones in the center of the fovea

Slide by Steve Seitz

2.7 컬러

- Electromagnetic spectrum



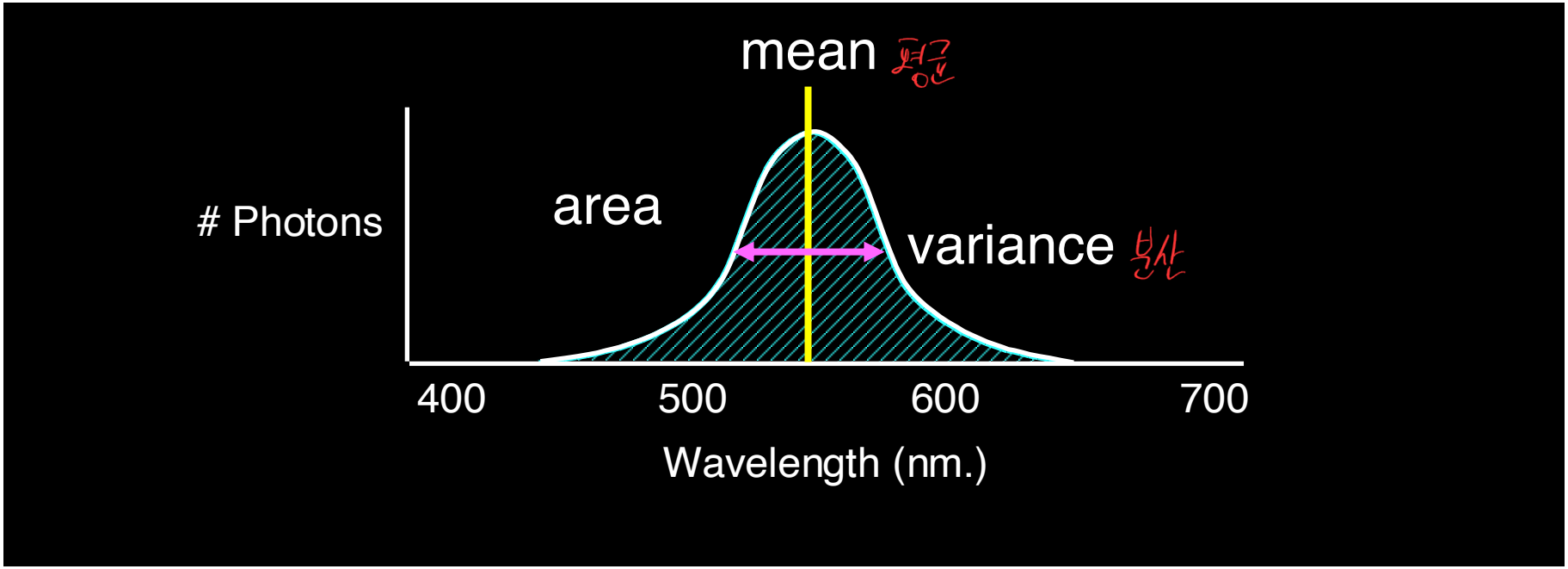
Human Luminance Sensitivity Function

2.7 컬러

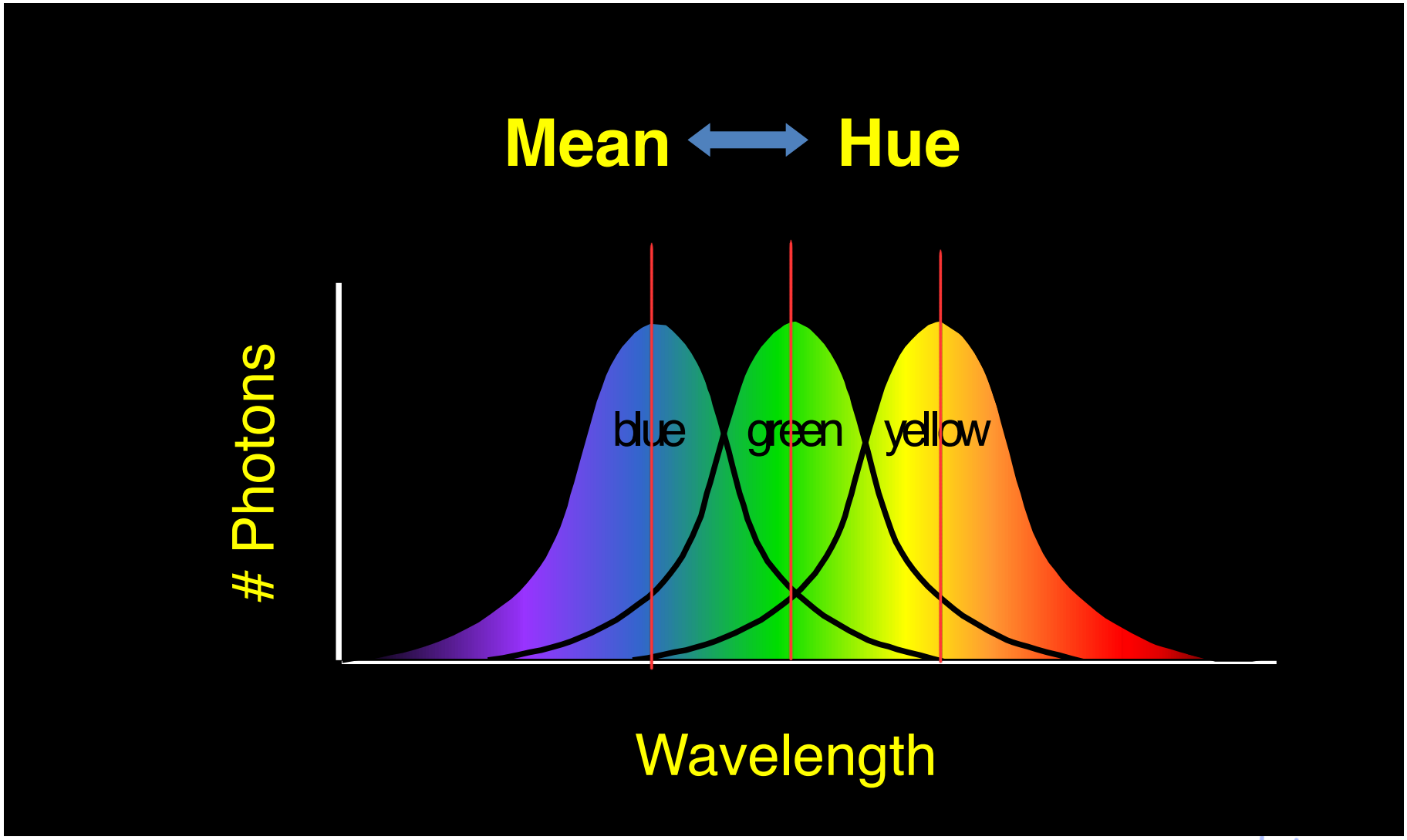
There is no simple functional description for the perceived color of all lights under all viewing conditions, but ……

A helpful constraint:

Consider only physical spectra with normal distributions

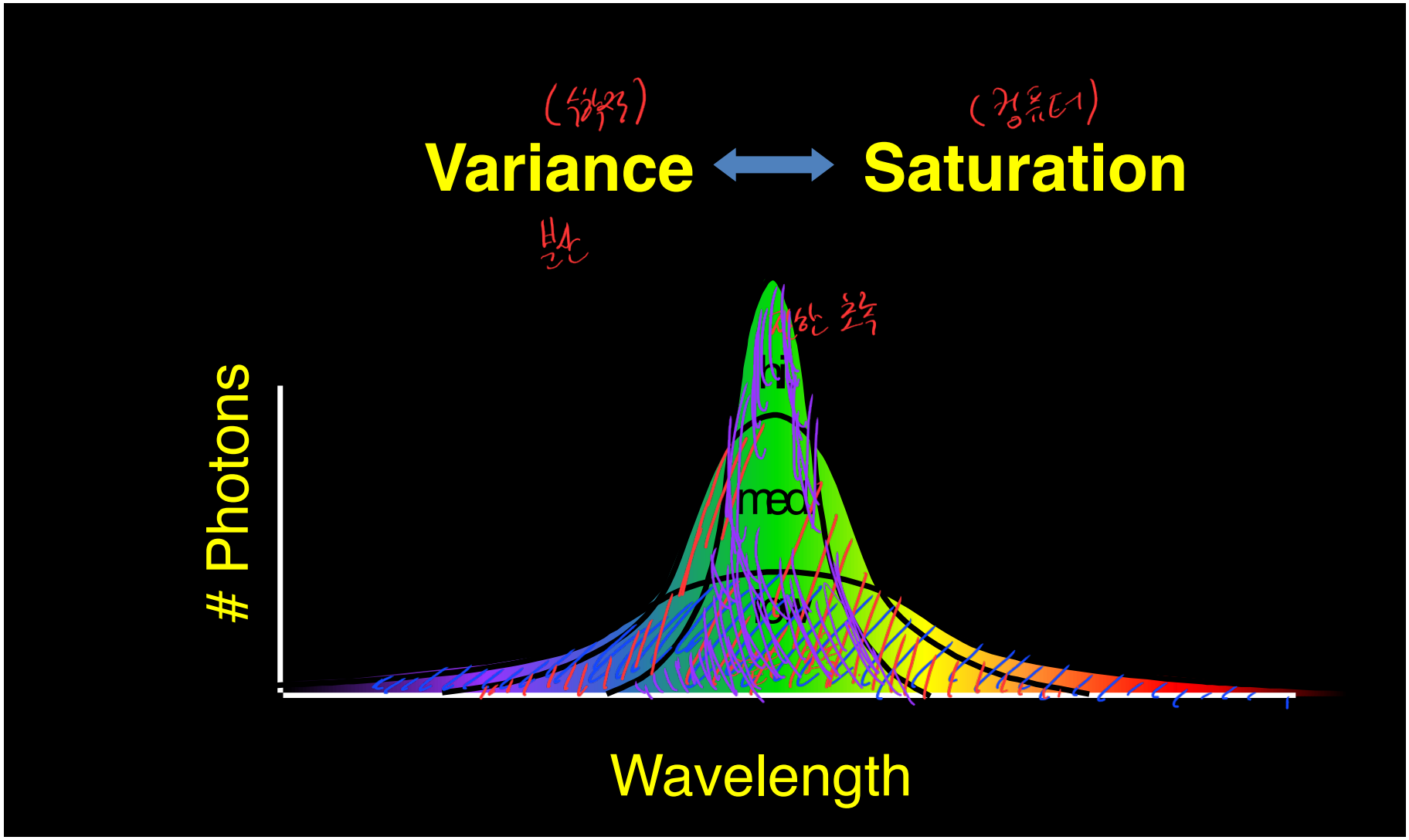


2.7 컬러



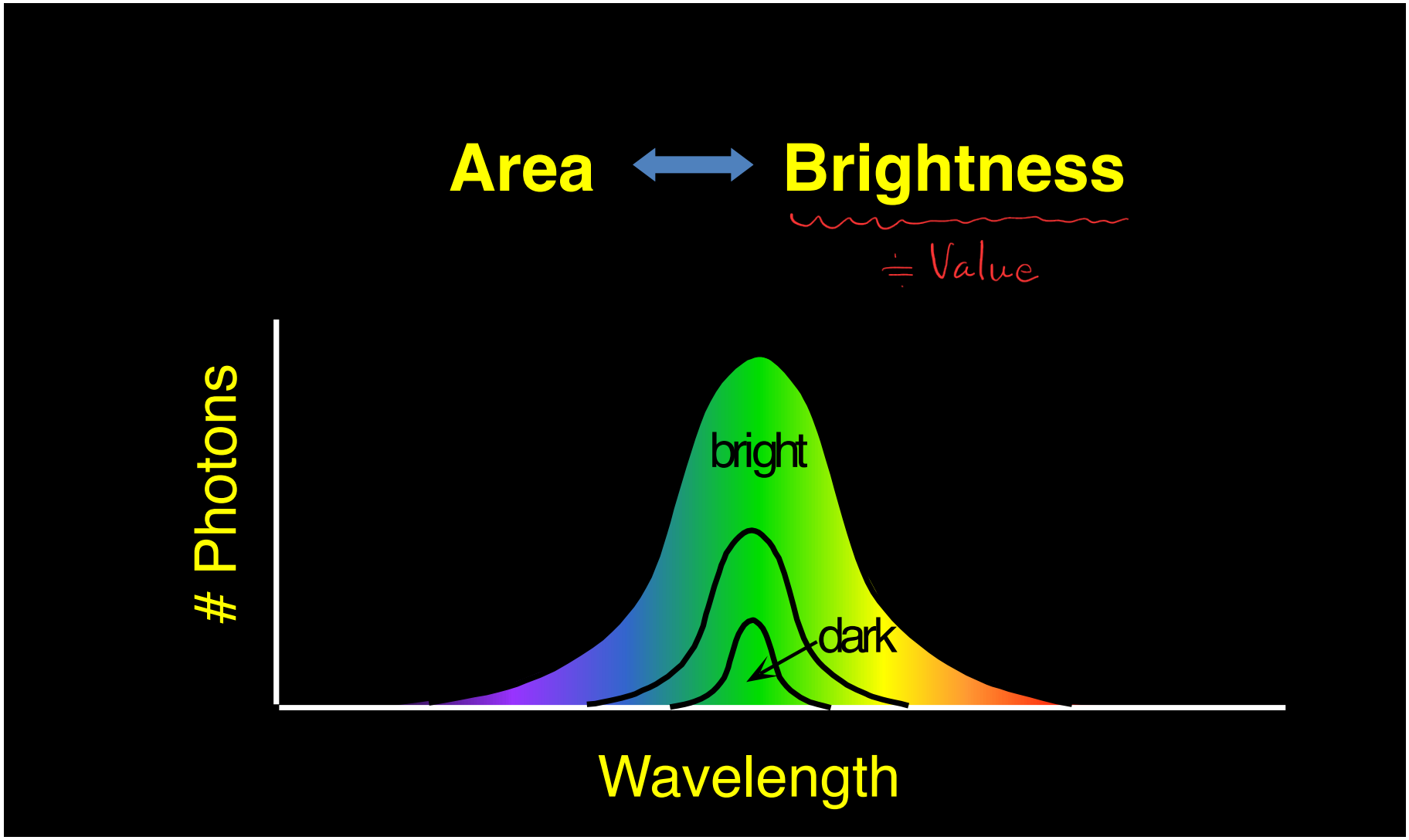
보라 파장 ← 가시광선 → 적색선 빨강

2.7 컬러



2.7 컬러

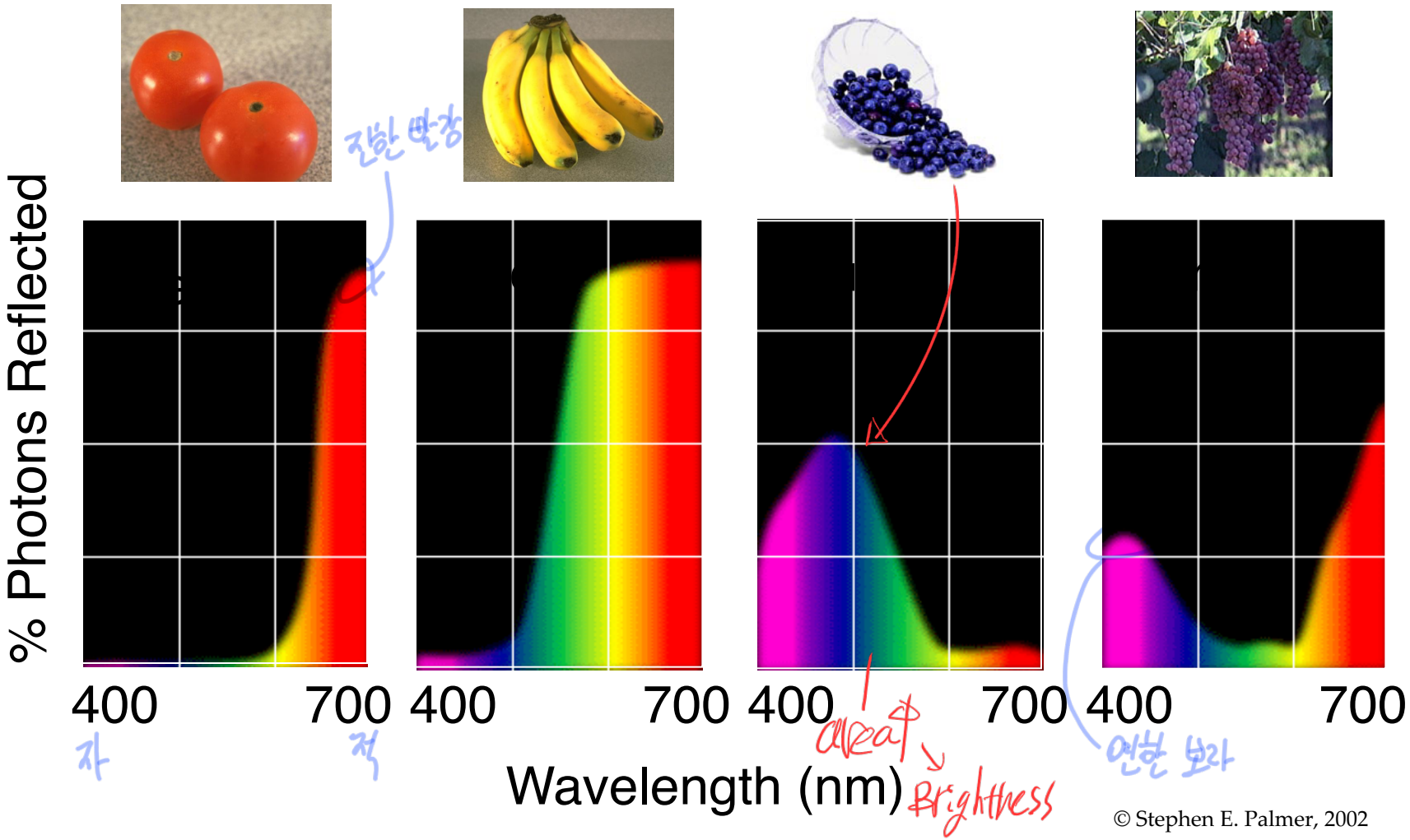
HSV



2.7 컬러

- The Physics of Light

Some examples of the reflectance spectra of surfaces

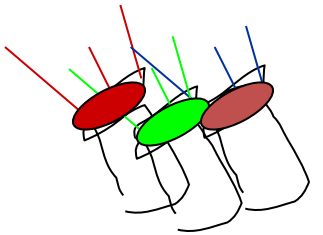


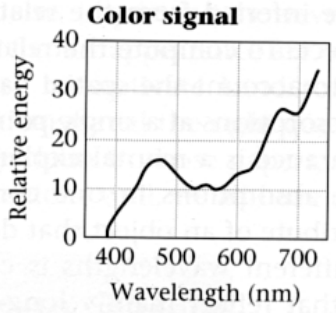
Linear color spaces RGB

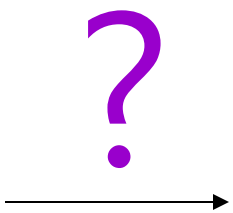
- How to compute the weights of the primaries to match any spectral signal?

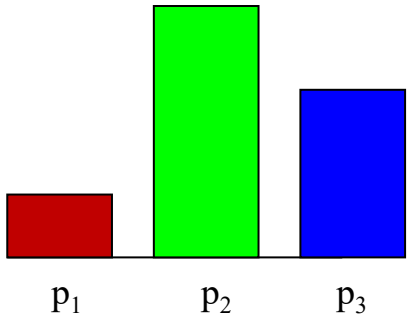
Given: a choice of three primaries and a target color signal

Find: weights of the primaries needed to match the color signal









Linear color spaces

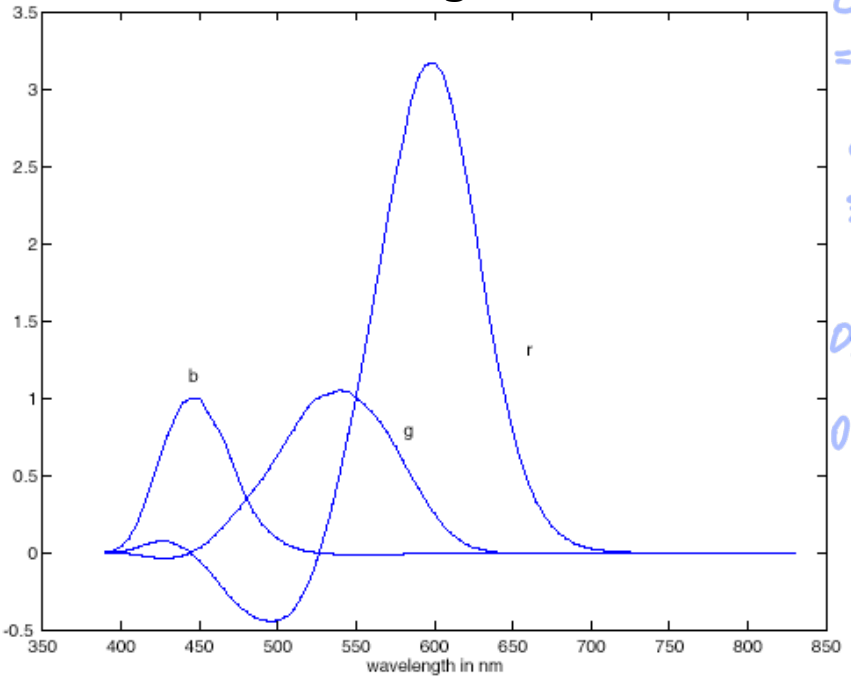
- In addition to primaries, need to specify **matching function s:** the amount of each primary needed to match a monochro matic light source at each wavelength

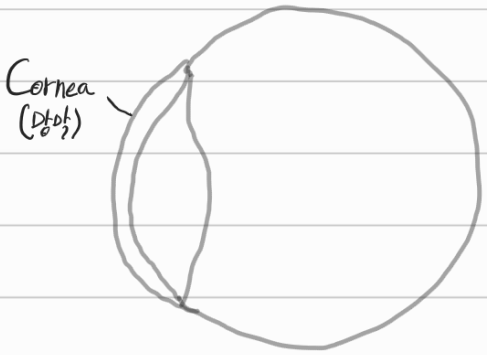
RGB primaries



$p_1 = 645.2$ nm
$p_2 = 525.3$ nm
$p_3 = 444.4$ nm

RGB matching functions





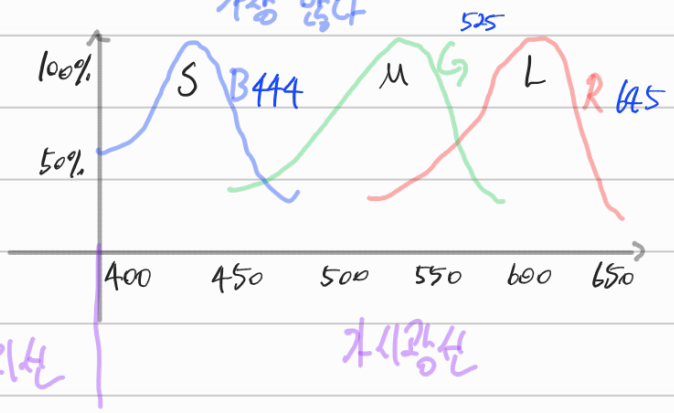
망막의 핵심 광수용체

Rod : 명암 감지

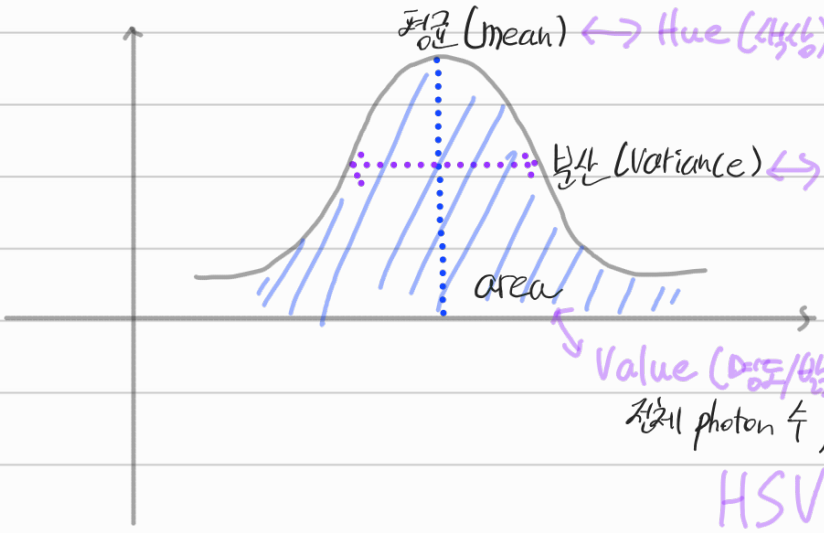
Cone : 색상 감지 , 파장 영역에 따라 S, M, L

R G B

가장 많다



자외선 | 가시광선 | 적외선

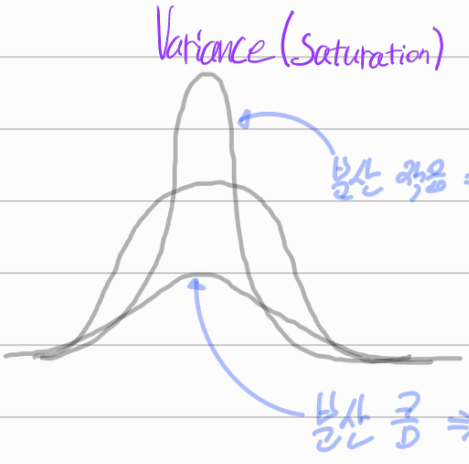


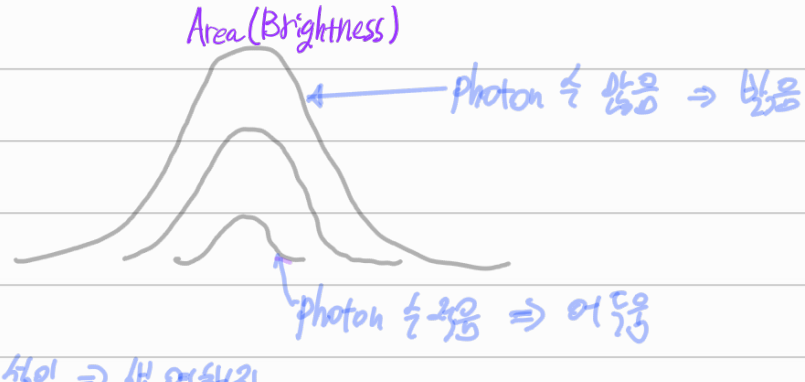
좁고 뾰족 ⇒ 특정 파장 빛만 ⇒ 채도↑, 선명

HSV

평균이 어느 위치에 형성되느냐에 따라



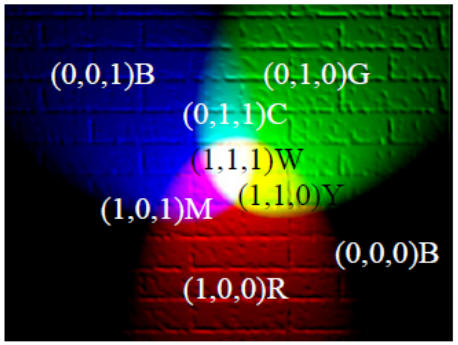




2.7 컬러

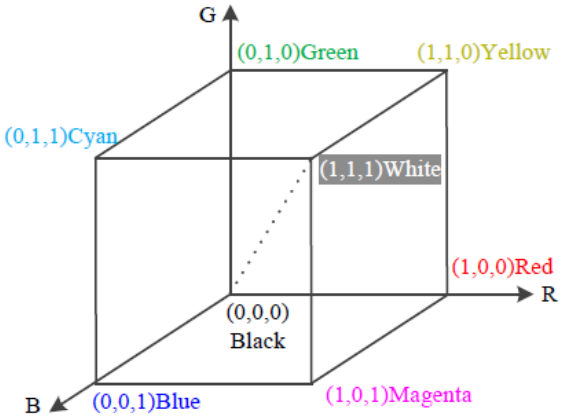
RGB 모델

- 길이가 1인 정육면체로 색을 표현

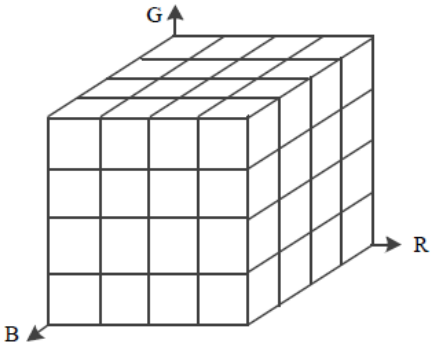


(1,0,0)=Red
(0,1,0)=Green
(0,0,1)=Blue
(0,1,1)=Cyan
(1,0,1)=Magenta
(1,1,0)=Yellow
(0,0,0)=Black
(1,1,1)=White

(a) 빛의 혼합



(b) RGB 큐브

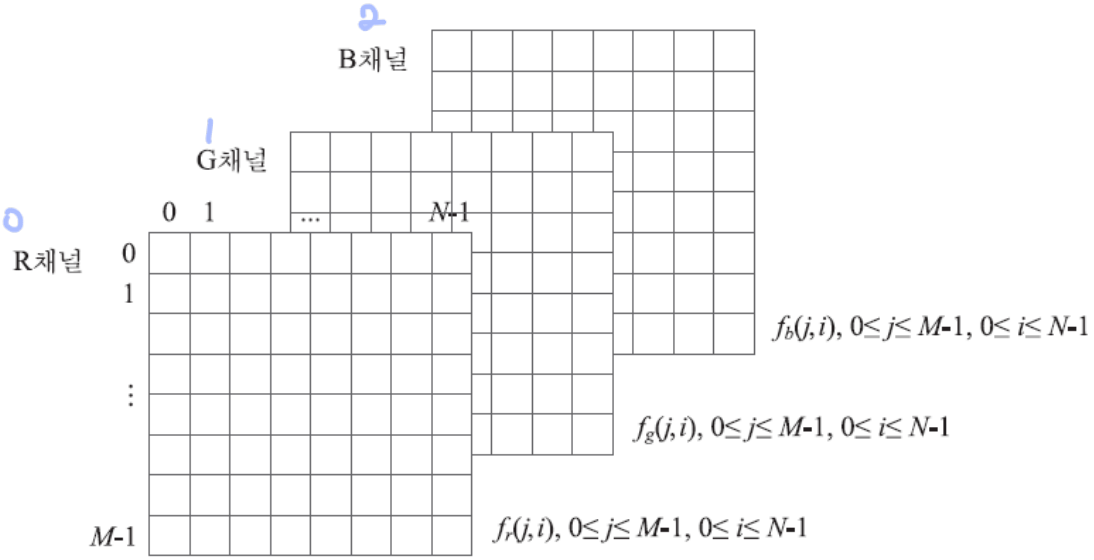


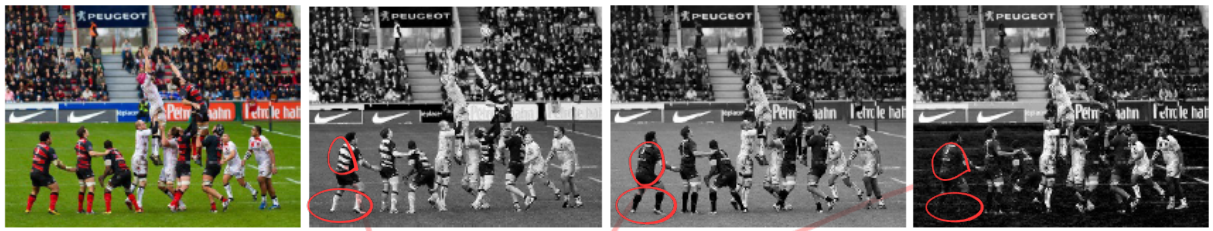
(c) 양자화된 4×4×4 RGB 큐브

그림 2-43 RGB 컬러 모델

2.7 컬러

- RGB 모델로 영상 표현
  - f_r, f_g, f_b의 세 채널로 표현





> RGB 영상(원래 영상) > f_r > f_g > f_b

그림 2-44 RGB 컬러 영상

```
for i = 1 : N-1
  for j = 1 : M-1
    for k = 0 : 2
      (i j k)

0 : R - channel
1 : G - "
2 : B - "
```

2.7 컬러

- HSI 모델
  - 이중 콘으로 색을 표현

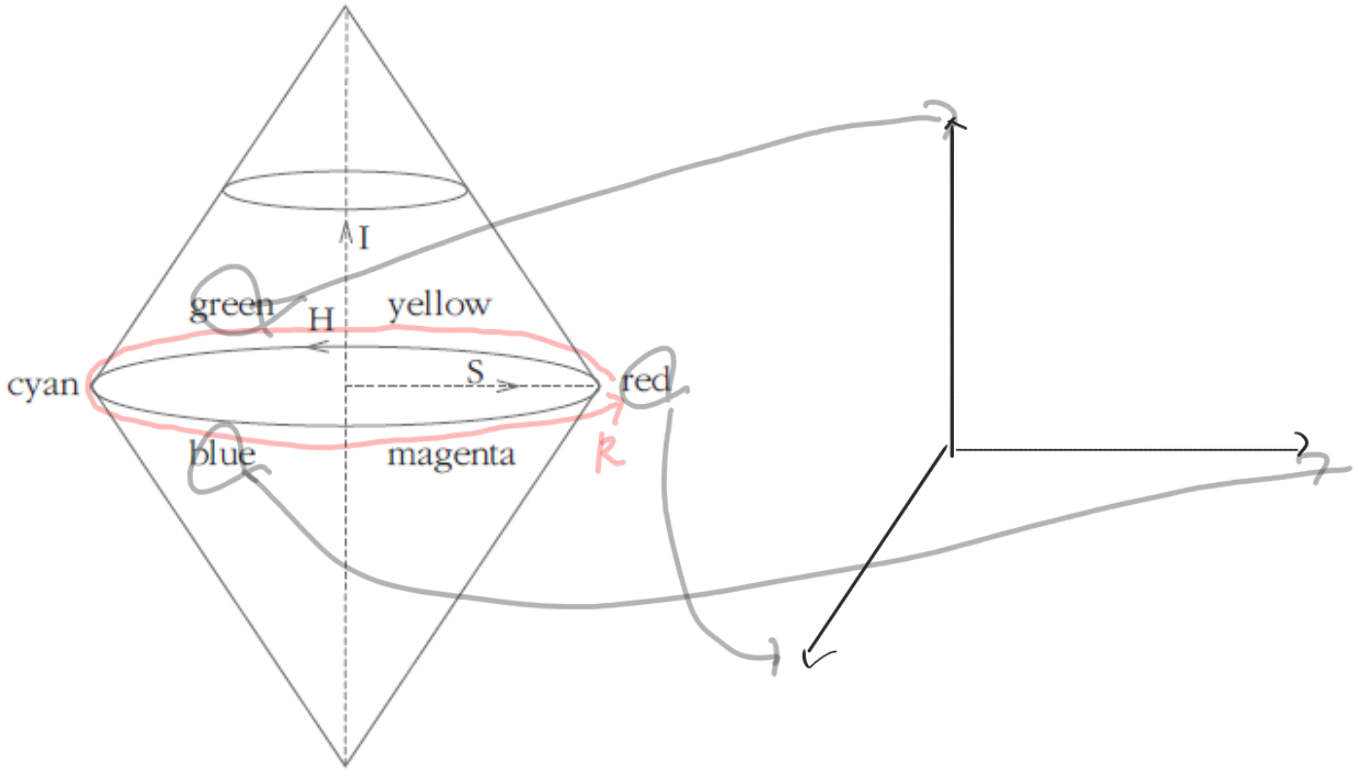
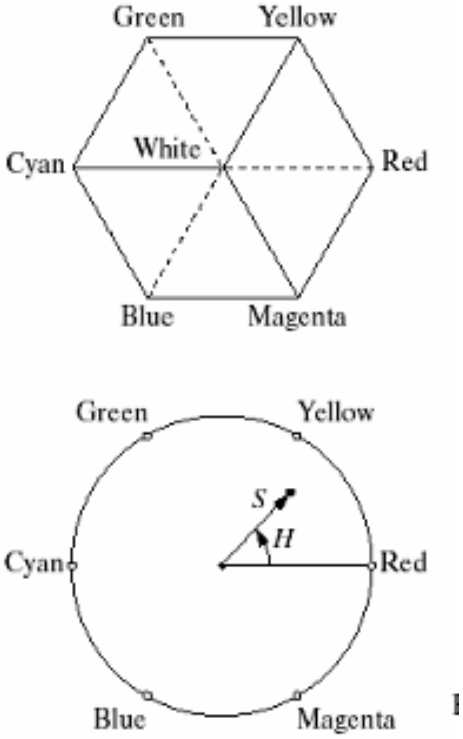
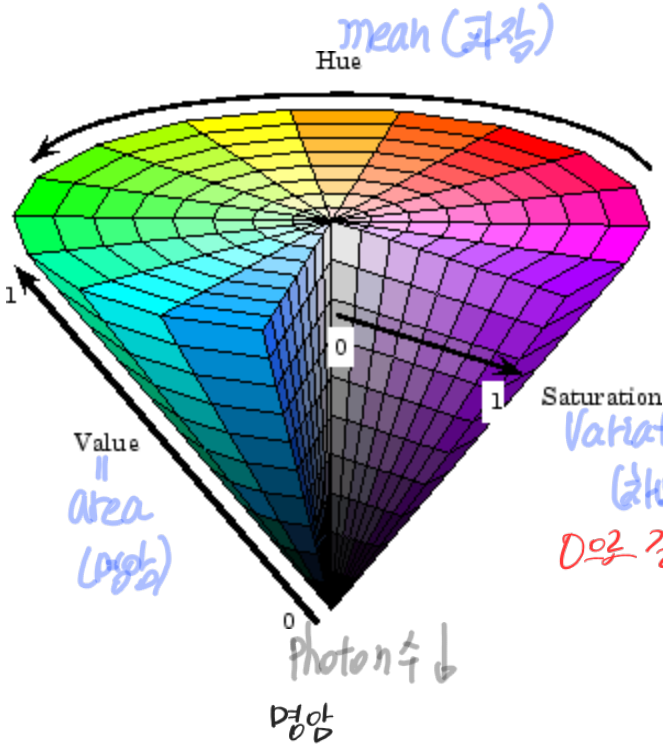
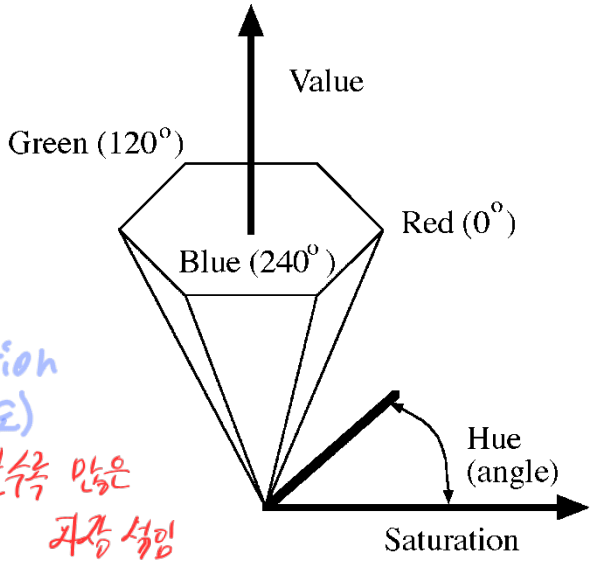


그림 2-45 HSI 컬러 모델

Nonlinear color spaces: HSV

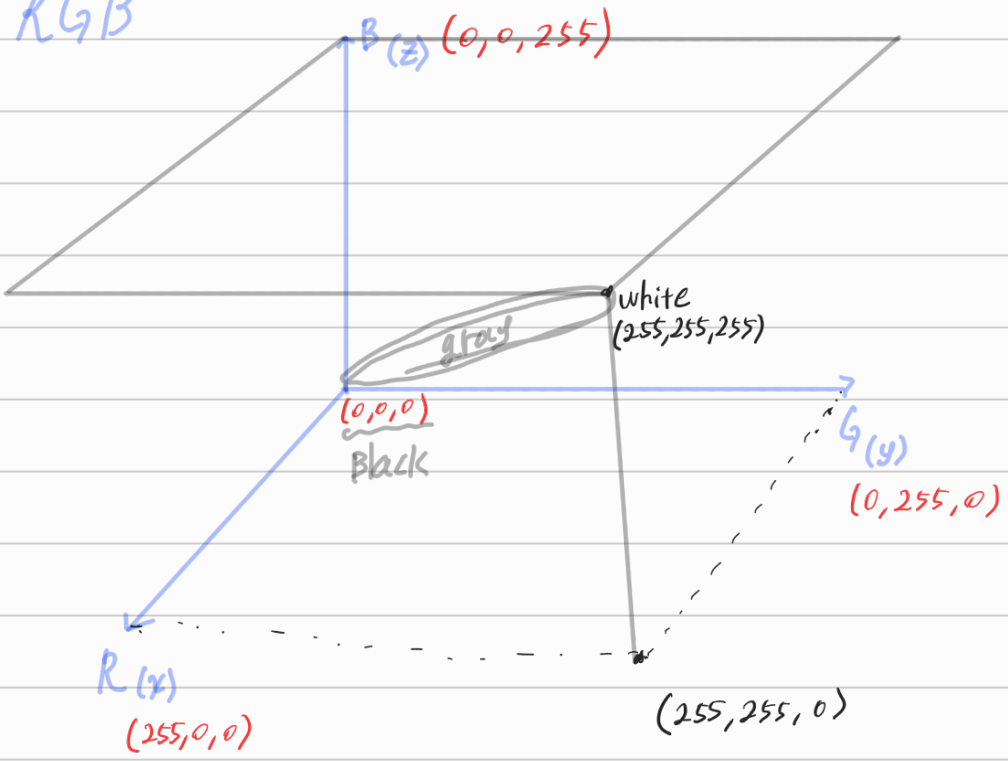






- Perceptually meaningful dimensions: Hue, Saturation, Value (Intensity)
- RGB cube on its vertex

RGB



Red = 1X255 + 0X255 + 0X255

모든 색상의 조합 : 256 X 256 X 256

Color = $x \cdot R + y \cdot G + z \cdot B$

$0 \le R, G, B \le 255$

$0 \le x, y, z \le 1$

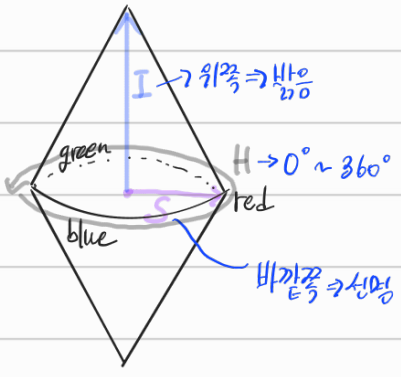
for k = 0 : 2 (0 : Red, 1 : Green, 2 : blue)

for i = 0 : N-1

for j = 0 : M-1

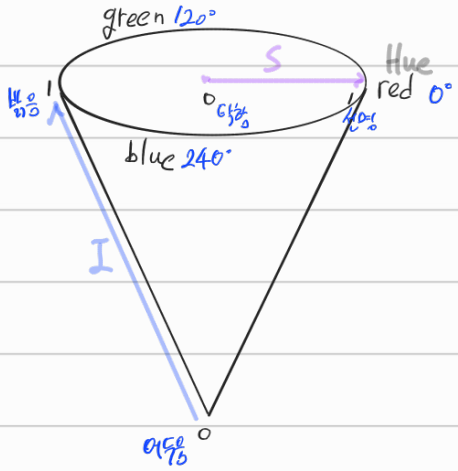
HSI 모델

: 이중 콘으로 색 표현



HSV 모델

: 비선형 색상 공간



Takeaut 3가지

1. Image → Python (라이브러리 이용)
2. Color (RGB, HSV)
3. Color → 정서적인 특징, 전체적인 값 X → 패턴 인식에 사용 X

2.7 컬러

- 컬러 영상 처리
  - 가장 단순한 방법은 세 채널을 독립적으로 처리



> RGB 영상 > f_r > f_g > f_b

그림 2-46 RGB 영상에 가우시안 스무딩(σ=2.0)을 적용한 결과

2.7 컬러

컬러 영상 처리



> 원래 영상 > 밝게(a=32) > 어둡게(a=32)

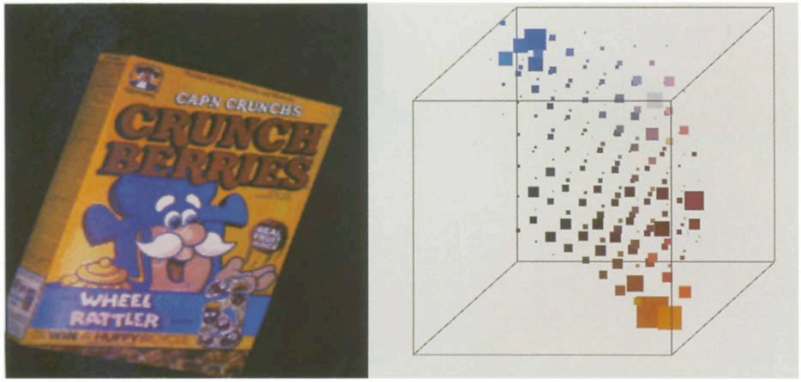
> 원래 영상 > 감마 수정(γ=0.67) > 감마 수정(γ=1.5)

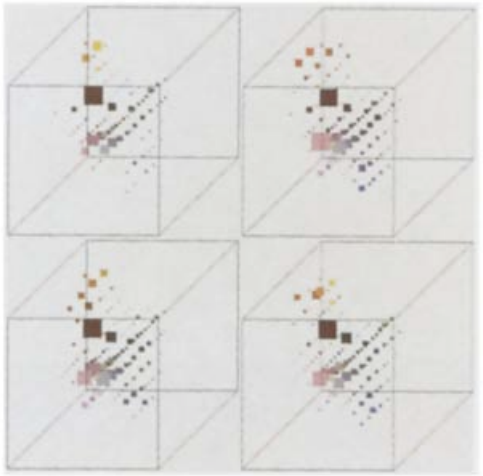
그림 2-47 RGB 영상에 점 연산을 적용한 결과

Uses of color in computer vision

Color histograms for image matching







상표인식
ex) 상품 보고 상품 인식해서
자동 요리해

Swain and Ballard, Color Indexing, IJCV 1991.

Uses of color in computer vision

Segmentation
↳ SAM3

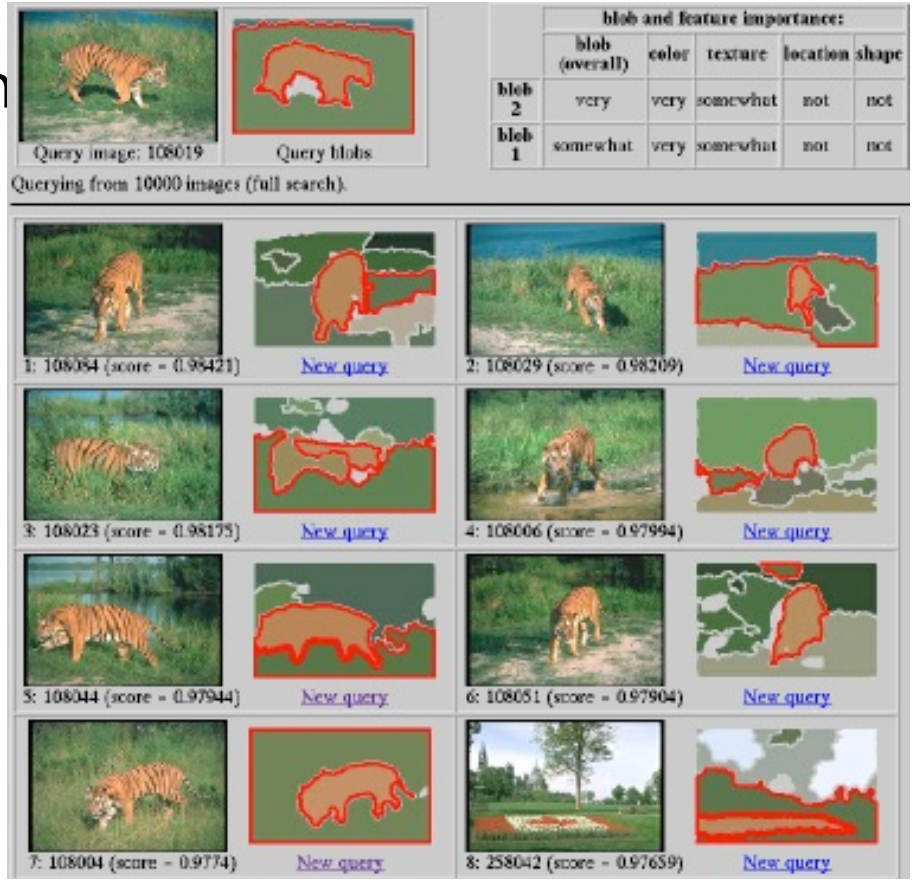
idée **Multicolr Search Lab**



http://labs.ideeinc.com/multicolr

Uses of color in computer vision

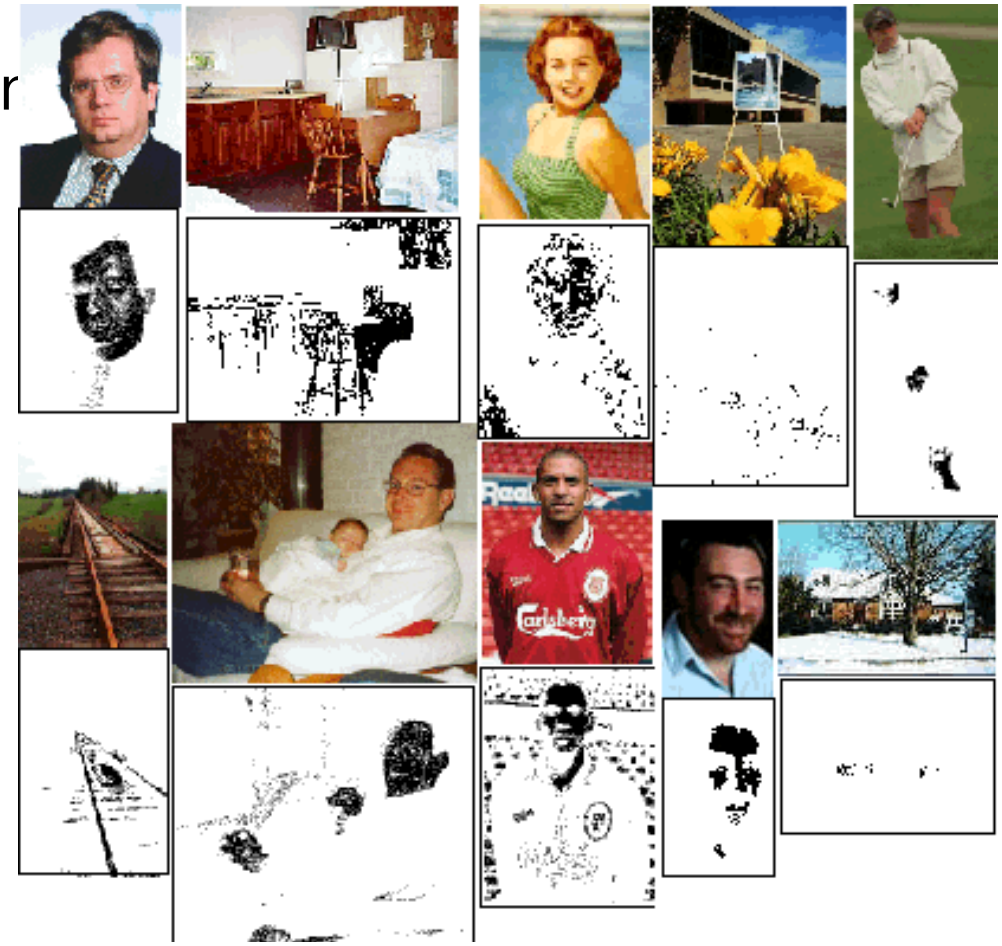
Image segmen



C. Carson, S. Belongie, H. Greenspan, and Ji. Malik, Blobworld: Imag e segmentation using Expectation-Maximization and its application to image querying, ICVIS 1999.

Uses of color in computer vision

Skin detection



M. Jones and J. Rehg, Statistical Color Models with Application to Skin Detection, IJCV 2002.

Uses of color in computer vision

Robot soccer



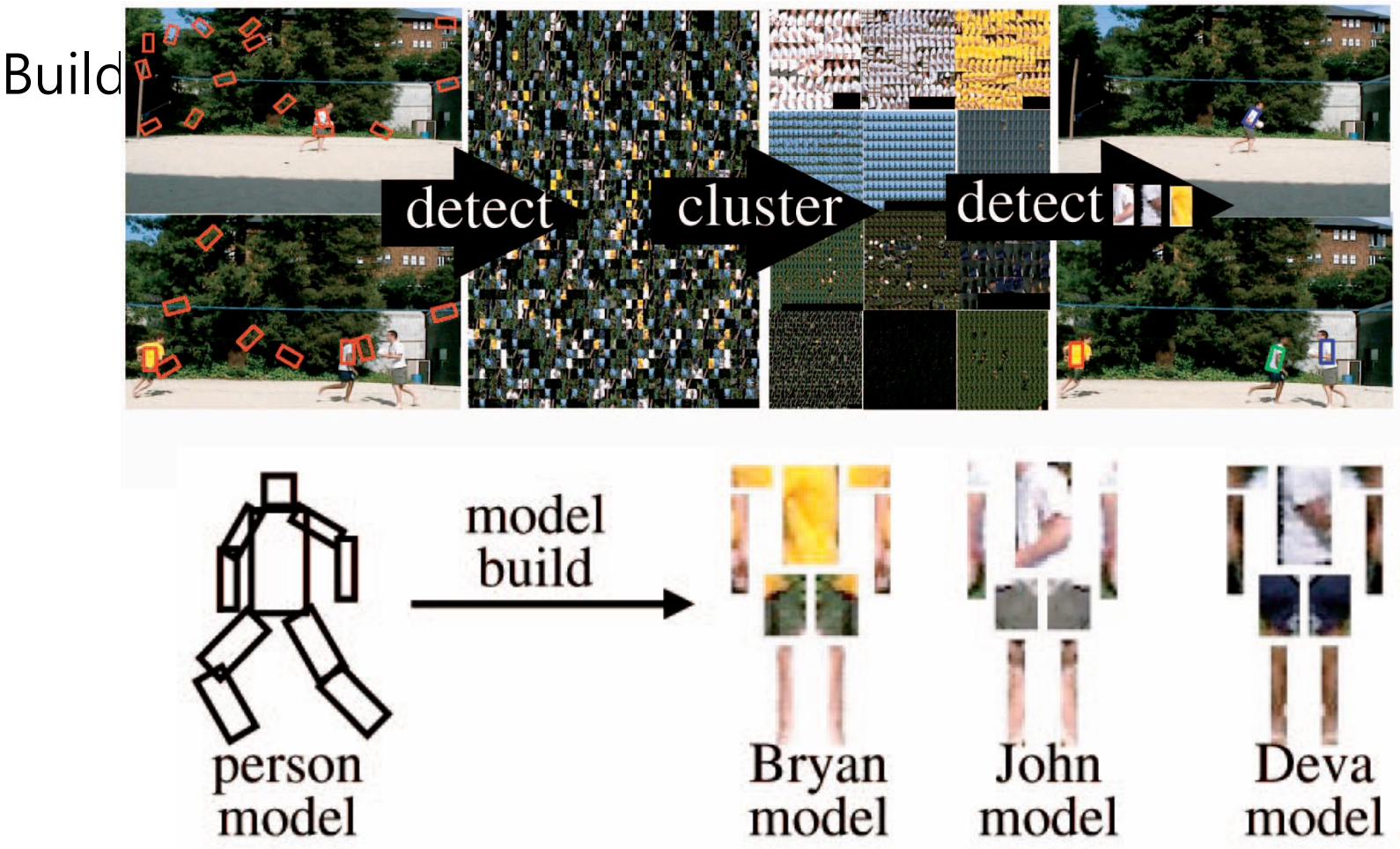
M. Sridharan and P. Stone, Towards Eliminating Manual Color Calibration at RoboCup. RoboCup-2005: Robot Soccer World Cup IX, Springer Verlag, 2006

Takeaut 3가지

1. Image → Python (라이브러리 이용)
2. Color (RGB, HSV)
3. Color → 정서적인 특징, 절대적인 값 X → 패턴 인식에 사용 X

Source: K. Grauman

Uses of color in computer vision



D. Ramanan, D. Forsyth, and A. Zisserman. Tracking People by Learning their Appearance. PAMI 2007.